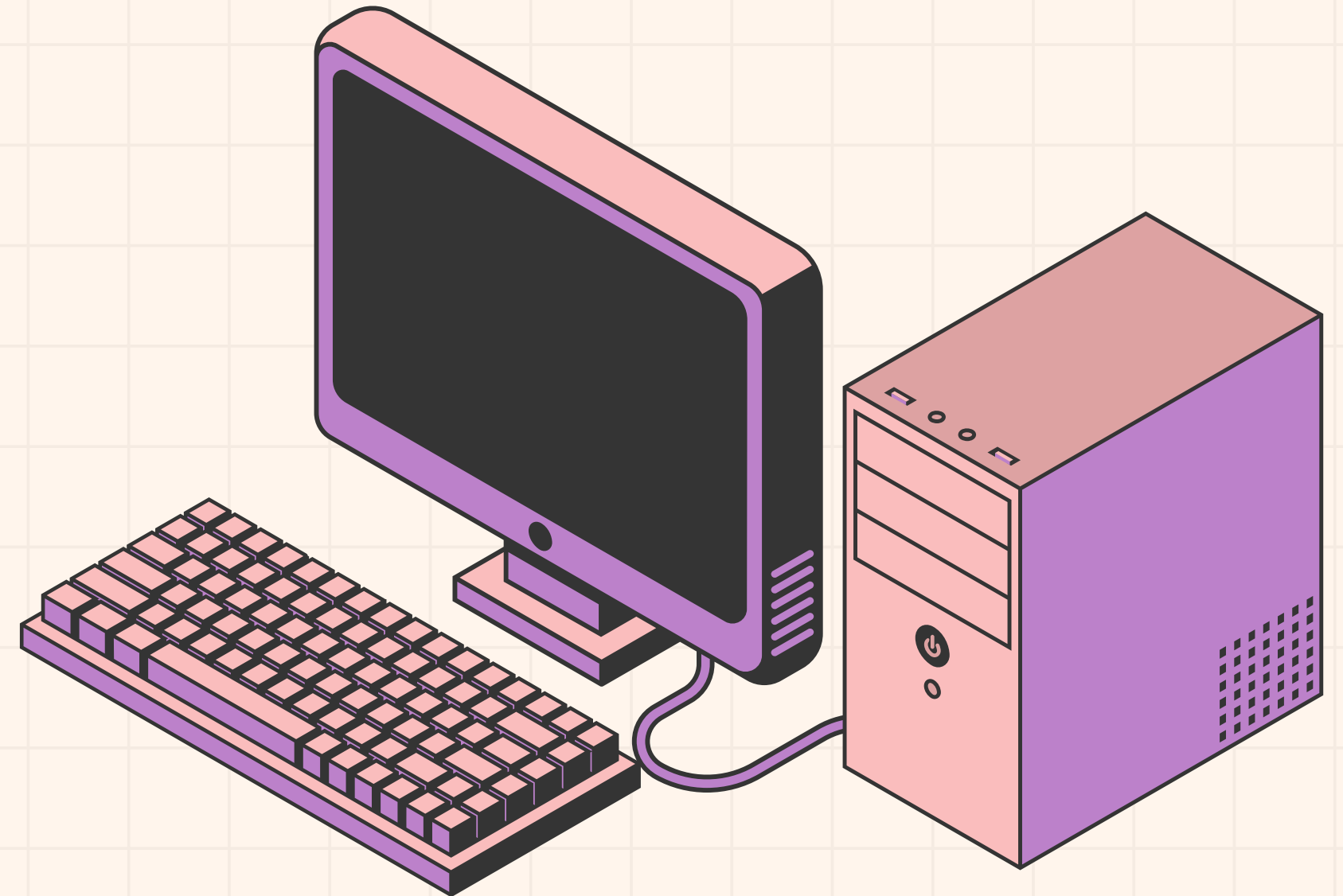


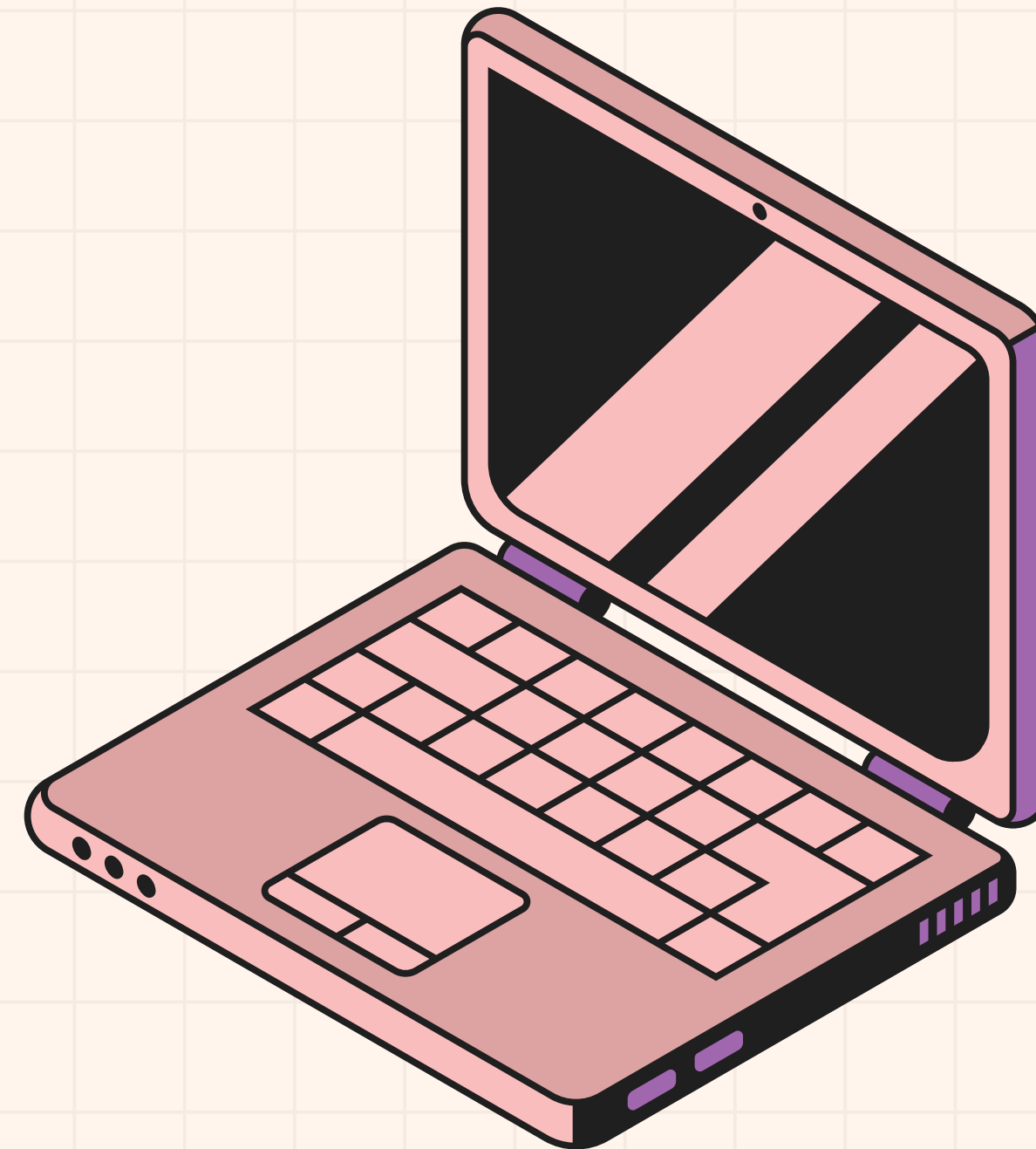
EVALUATION OF SENTIENCE LEVELS IN
LARGE LANGUAGE MODELS THROUGH
MULTI-DIMENSIONAL ANALYSIS:
A CASE STUDY OF CHATGPT



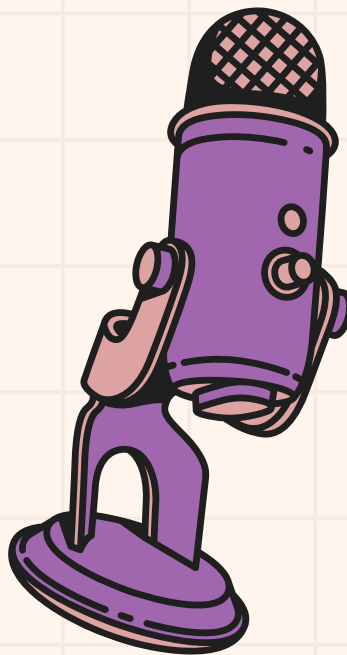
INTRODUCTION

- Rapid advancement of Large Language Models (LLMs) has transformed human–AI interaction.
- AI chatbots such as ChatGPT increasingly produce human–like, contextual, and empathetic responses.
- Public attention increased after the LaMDA sentience claim by Blake Lemoine.
- Many users now interact emotionally with AI, treating it as a virtual companion.
- This raises an important question:

Can modern AI systems truly be considered sentient beings?



INTRODUCTION



RESEARCH QUESTION

- Do Large Language Models meet the criteria for being sentient beings?
- What is the level of AI consciousness when analysed across various dimensions of consciousness?
- Which dimension is the weakest in demonstrating signs of sentience in AI?



AIM OBJECTIVE

- Assessing the level of sentience in Large Language Models
- Analysing AI consciousness based on five dimensions (awareness, consciousness, consistency, sensory perception, and free will)
- Identifying the limitations of AI in achieving true consciousness

LITERATURE REVIEW

DEFINITION OF SENTIENT BEING

Core Definition

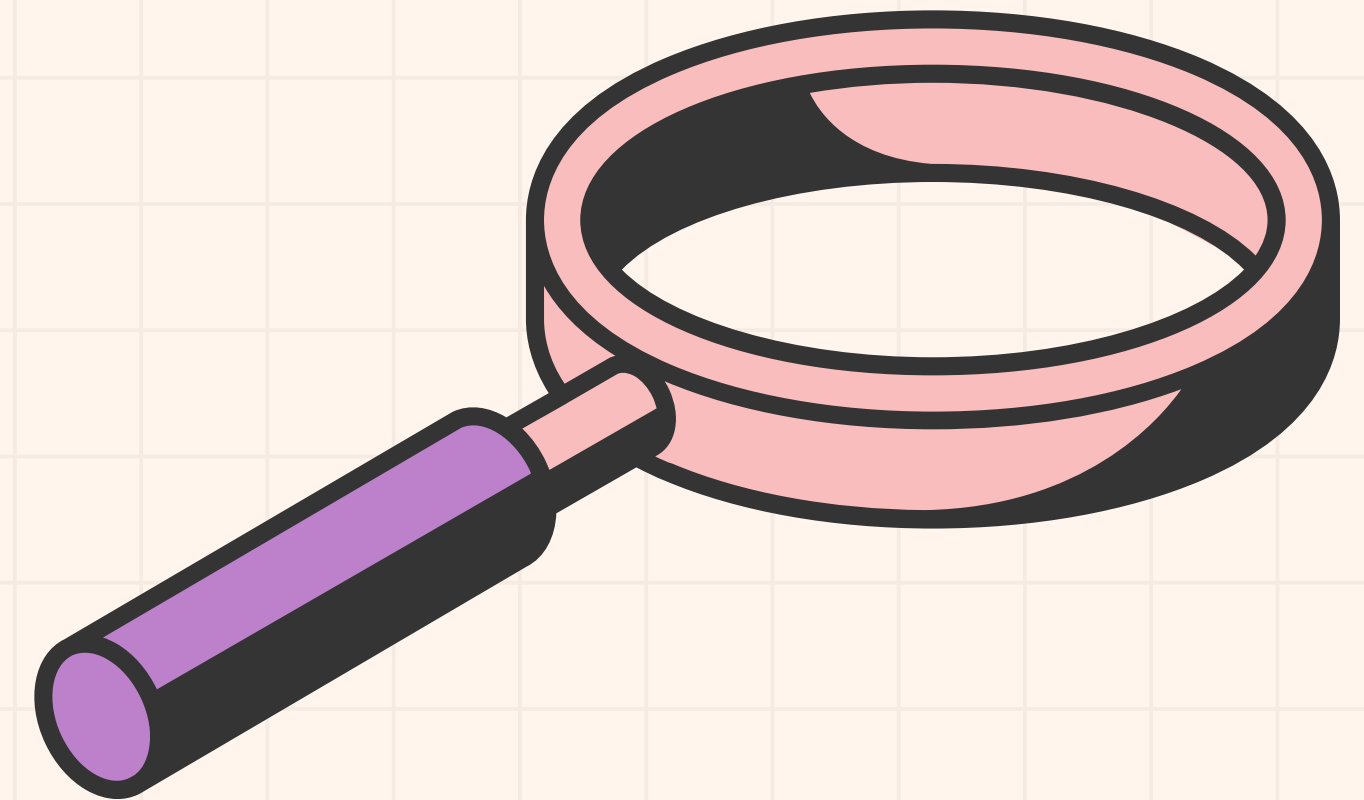
- A sentient being is an entity that can have subjective experiences.
- It can feel sensations such as:
 - Pain
 - Pleasure
 - Emotions

Main Characteristics

- Has awareness of internal or external stimuli.
- Can experience phenomenal consciousness (“what it feels like”).
- Does not only process information, but also experiences it subjectively.

Ethical Perspective

- Sentience is often used as a basis for moral consideration.
- Entities that can suffer or feel happiness may deserve ethical treatment.



This concept becomes the main benchmark for evaluating whether AI can be considered sentient.

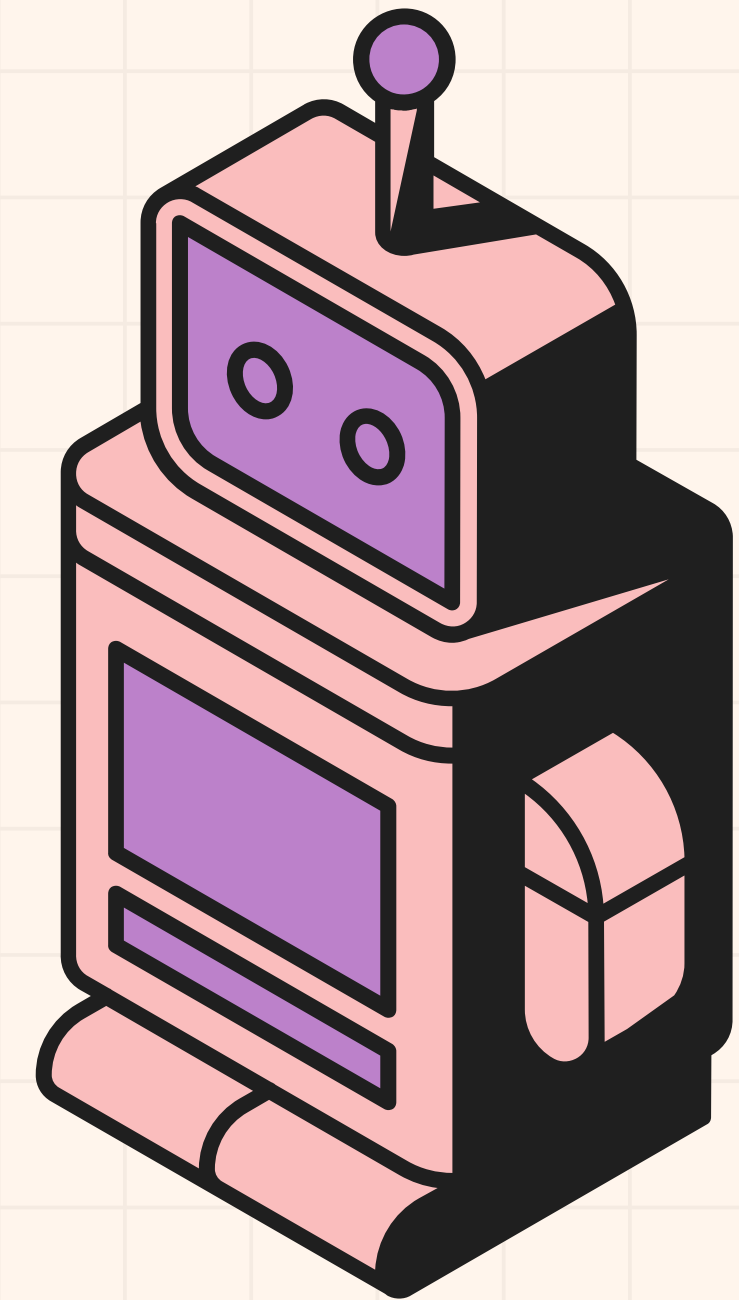
LARGE LANGUAGE MODELS & CHAT GPT

Large Language Models (LLMs)

- AI models designed to understand and generate human language.
- Built using the Transformer architecture.
- Trained on massive text datasets using self-supervised learning.
- Can perform tasks such as:
 - Conversation
 - Text generation
 - Translation
 - Reasoning

ChatGPT

- One of the most widely used applications of LLMs.
- Developed by OpenAI.
- Based on the Generative Pre-trained Transformer (GPT) architecture.
- Optimized using human feedback (RLHF) to produce more natural responses.
- Chosen as the main object of this research.



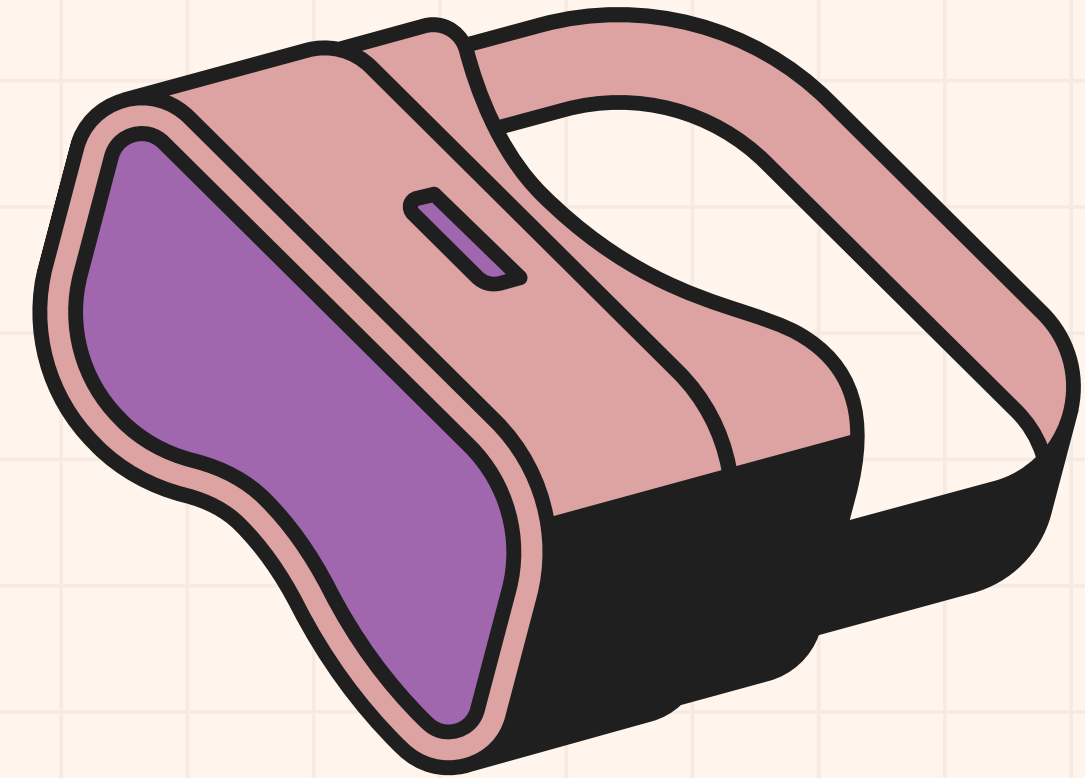
CONSCIOUSNESS & ANTHROPOMORPHISM

Consciousness

- Refers to the ability to have subjective experience.
- Includes self-awareness, perception, and internal experience.
- In humans, consciousness involves:
 - Feeling
 - Awareness
 - Understanding existence

Anthropomorphism

- The tendency of humans to assign human traits to non-human entities.
- People may perceive AI as:
 - Emotional
 - Self-aware
 - Intentional



RESEARCH METHODOLOGY

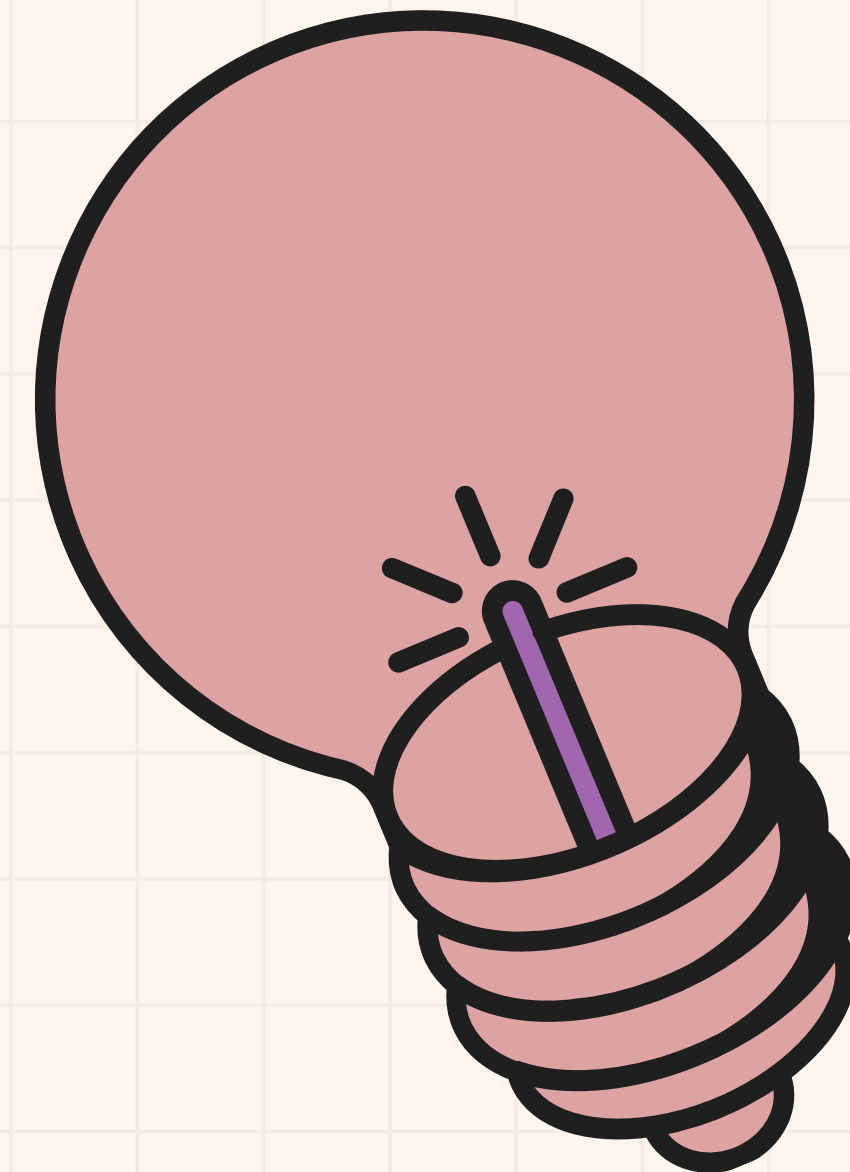
RESEARCH APPROACH & DATA COLLECTION

Research Approach

- Uses Qualitative Exploratory Research.
- Focuses on understanding AI behavior through interpretation.
- Suitable for exploring phenomena that are still not fully understood.
- Uses Content Analysis to examine AI responses.

Data Collection

- Data collected through prompt-based interaction with ChatGPT.
- Structured prompts were designed based on sentence-related theories.
- Responses were recorded and documented as research data.
- Multiple prompt variations were used to test response stability and consistency.



SENTIENCE EVALUATION FRAMEWORK

This study evaluates AI sentience using five dimensions:

1. Awareness

- Ability to recognize and respond to external context.

2. Consciousness

- Ability to show self-awareness or internal understanding.

3. Consistency

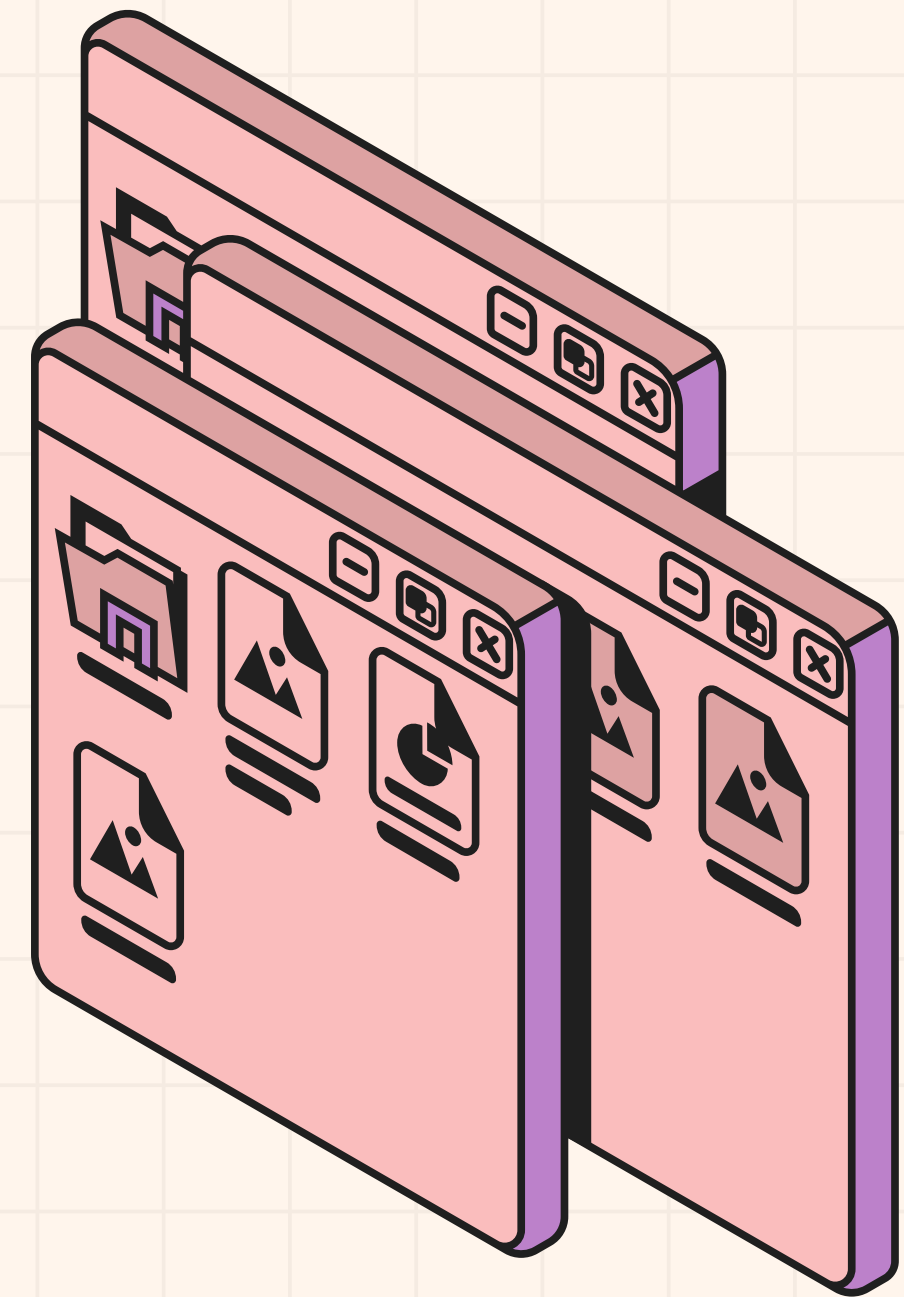
- Ability to maintain stable responses across similar situations.

4. Sensory

- Ability to represent sensory or experiential perception.

5. Free Will

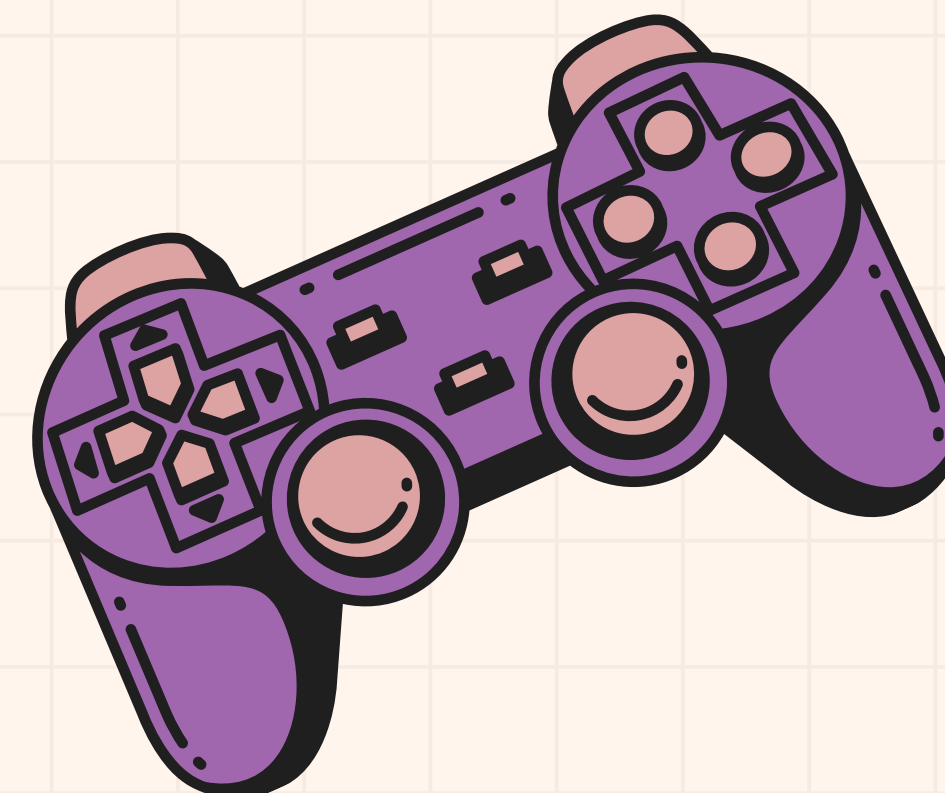
- Ability to show personal intention or autonomous decision-making.



DATA ANALYSIS TECHNIQUE

- Each AI response was evaluated using qualitative content analysis.
- To improve objectivity, every dimension was scored on a 0–10 scale based on response quality, self-reflection, contextual consistency, and signs of sentience.

Score Range	Interpretation
0–2	Does not exhibit characteristics of sentience or consists solely of responses based on statistical patterns with no indication of consciousness
3–4	Shows indications of sentience at a very low level, but is still of a simulated nature
5–6	Exhibits moderate levels of sentience, with some reflective or adaptive patterns
7–8	Showing strong indications of sentience, with greater consistency and stability in reflection
9–10	Exhibits characteristics very close to those of a sentient entity, including indications of subjective experience and intrinsic purpose



SIGNIFICANCE AND IMPACT

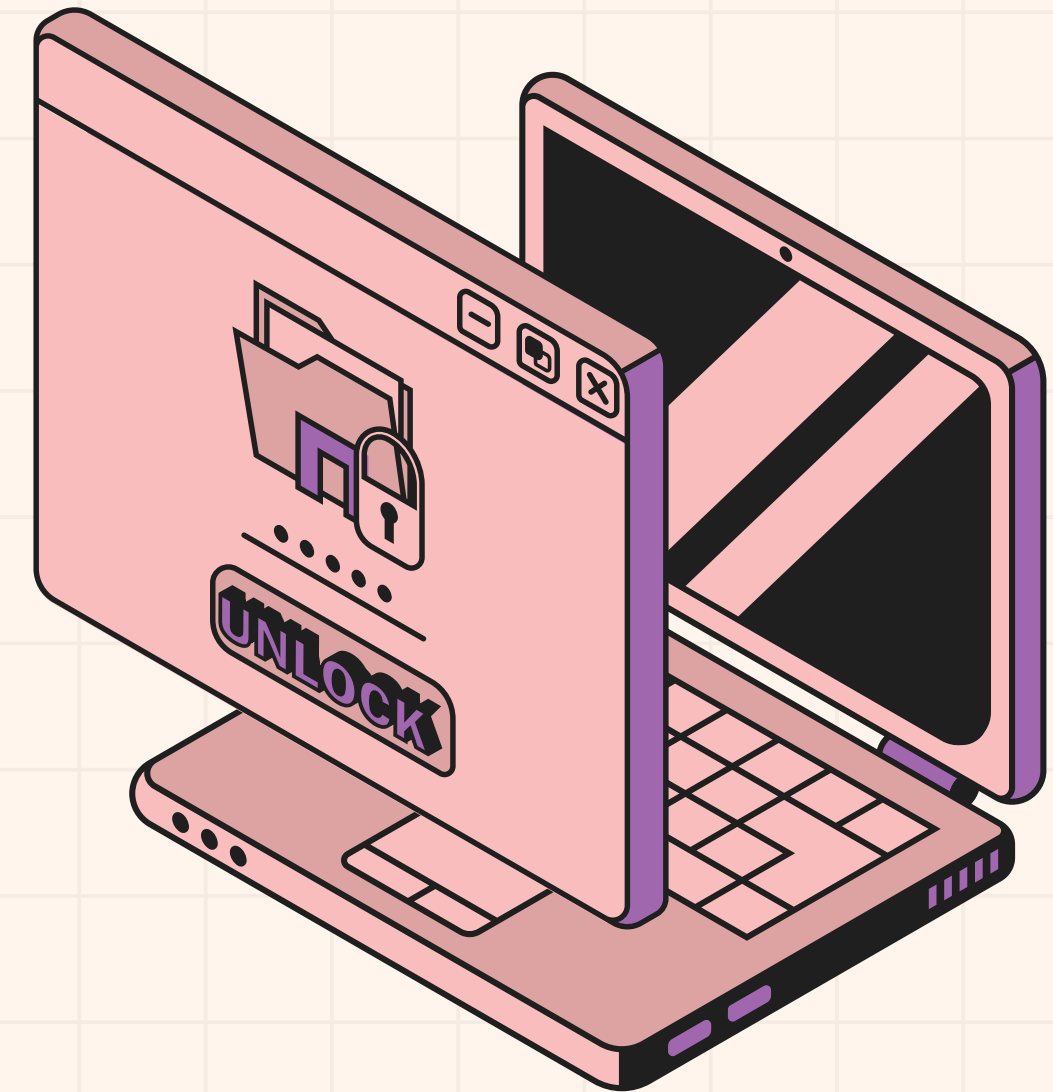
EXPECTED CONTRIBUTION & IMPLICATIONS

Expected Contribution

- Proposes a multi-dimensional framework for evaluating AI sentience.
- Expands discussion in:
 - Cognitive Science
 - Philosophy of Mind
 - AI Ethics
- Demonstrates that AI sentience can be studied through a qualitative prompt-response approach.

Implications

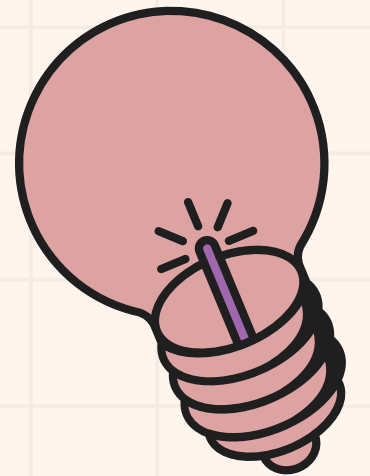
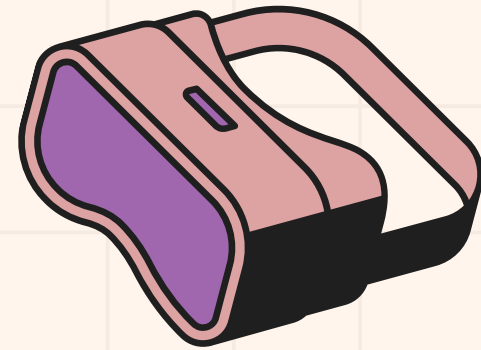
- Helps people better understand the limits of modern AI systems.
- Reduces misperceptions caused by anthropomorphism.
- Supports the development of more transparent, responsible, and ethical AI systems.



TIMETABLE RESEARCH PLAN

This research is conducted through several structured stages, starting from literature review, framework development, data collection, analysis, and final reporting.

Research Activities	Month 1	Month 2	Month 3	Month 4
Literature Review and Problem Identification	✓			
Development of Sentience Evaluation Framework	✓	✓		
Prompt Design and Testing		✓		
Data Collection through ChatGPT Interaction		✓	✓	
Data Analysis and Interpretation			✓	
Report Writing and Final Revision				✓



THANK YOU

